Nov 24, 2025

Selected Topics in Computational Quantum Physics

量子物理计算方法选讲

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Tensor networks

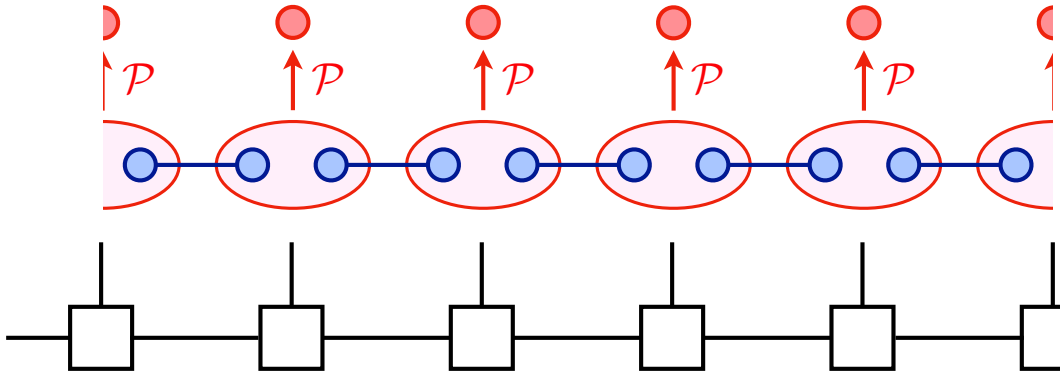
- Projected Entangled-Pair States (PEPS) and topological order
- 2D tensor network algorithms
- Multi-scale Entanglement Renormalization Ansatz (MERA)

Selected review articles:
F. Verstraete, J.I. Cirac, and V. Murg, arXiv: 0907.2796.
R. Orus, arXiv:1306.2164.
N. Schuch, arXiv:1306.5551.
J. C. Bridgeman and C. T. Chubb, arXiv:1603.03039.
R. Orus, arXiv: 1812.04011.
J.I. Cirac, D. Perez-Garcia, N. Schuch, F. Verstraete, arXiv:2011.12127.

New progress (update daily)
http://quattro.phys.sci.kobe-u.ac.jp/dmrg/condmat.html

Projected Entangled-Pair States (PEPS)

- matrix product states → 1D Projected Entangled-Pair States (PEPS)

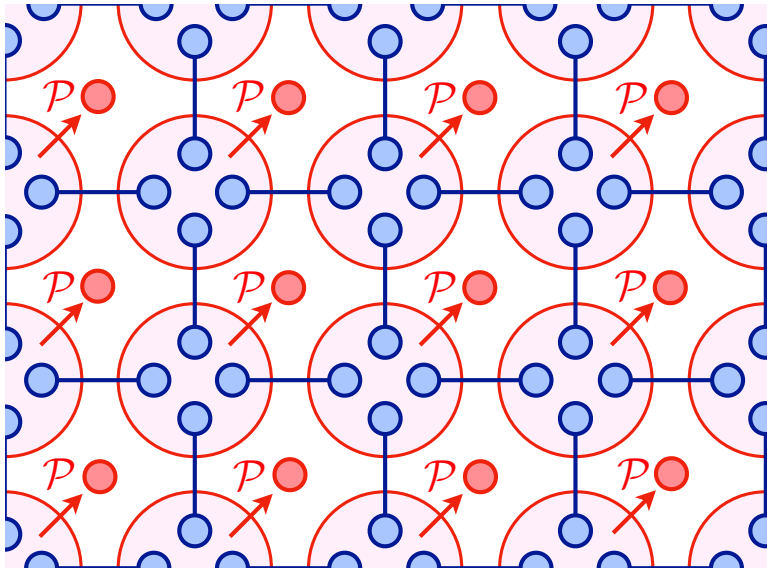


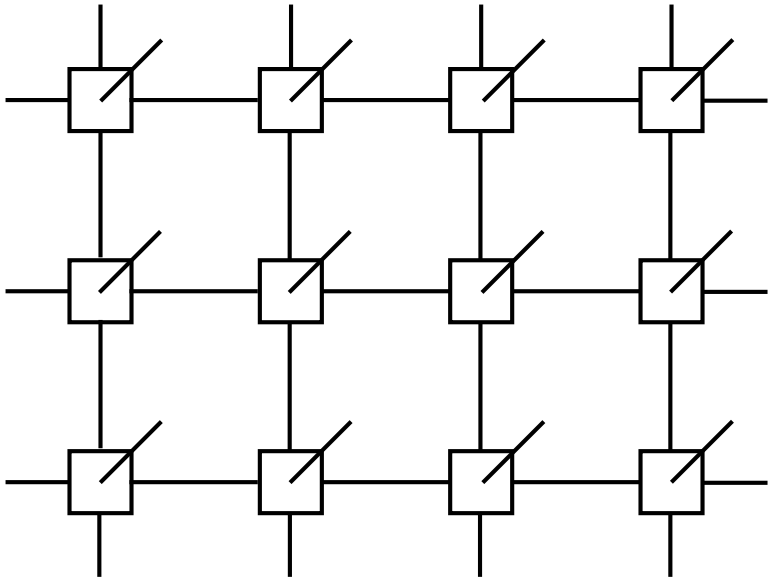


$$|\omega_D\rangle = \sum_{i=1}^{D} |i,i\rangle$$

$$|\Psi\rangle = \sum_{i_1,\ldots,i_N} c_{i_1\ldots i_N} |i_1,\ldots,i_N\rangle = (\mathcal{P}_1 \otimes \cdots \otimes \mathcal{P}_N)|\omega_D\rangle^{\otimes N} = \sum_{i_1,\ldots,i_N} \mathrm{tr}\left[A_{i_1}^{[1]} A_{i_2}^{[2]} \cdots A_{i_N}^{[N]}\right] |i_1,\ldots,i_N\rangle$$

- generate to 2D → 2D Projected Entangled-Pair States (PEPS)





States with exact PEPS form

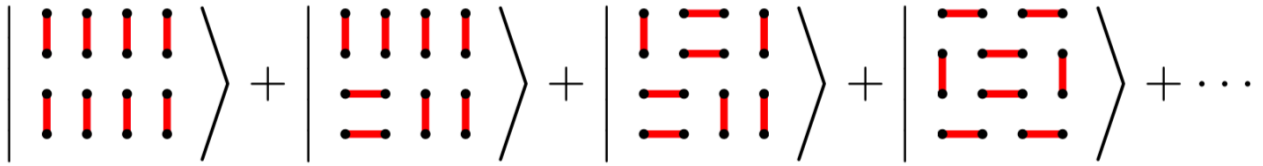
- product state $|\psi\rangle = (\alpha|0\rangle + \beta|1\rangle)^{\otimes N}$

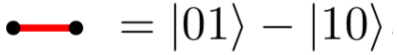
$$\mathcal{P} = \alpha|0\rangle\langle \begin{smallmatrix} & 0 & \\ 0 & & 0 \\ & 0 & \end{smallmatrix}| + \beta|1\rangle\langle \begin{smallmatrix} & 0 & \\ 0 & & 0 \\ & 0 & \end{smallmatrix}|, \qquad |\omega\rangle = |0,0\rangle$$

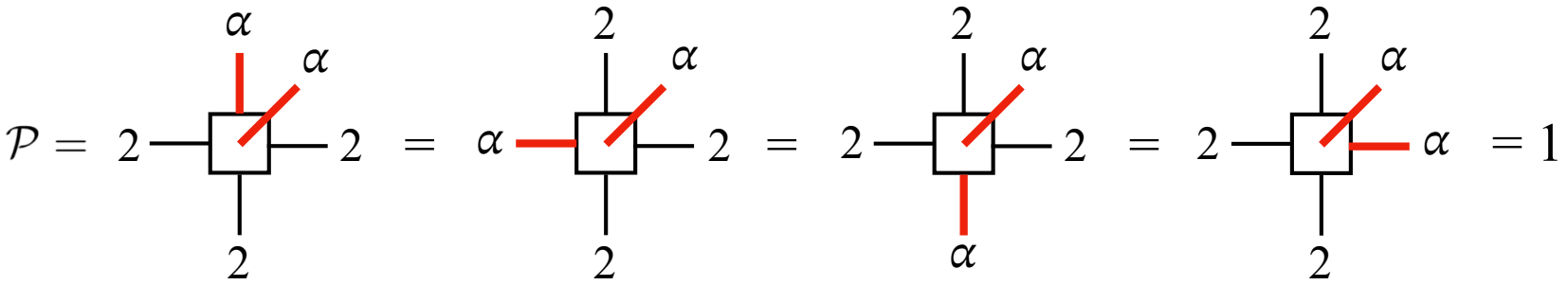
- GHZ state $|\psi\rangle = |00\cdots0\rangle + |11\cdots1\rangle$

$$\mathcal{P} = |0\rangle\langle \begin{smallmatrix} & 0 & \\ 0 & & 0 \\ & 0 & \end{smallmatrix}| + |1\rangle\langle \begin{smallmatrix} & 1 & \\ 1 & & 1 \\ & 1 & \end{smallmatrix}|, \qquad |\omega\rangle = |0,0\rangle + |1,1\rangle$$

- RVB state

$|\psi\rangle =$  $+ \cdots$





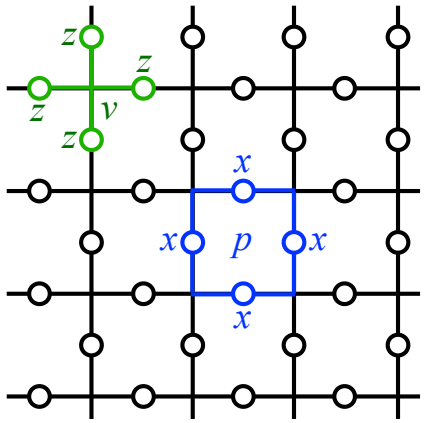
$|\omega\rangle = |0,1\rangle - |1,0\rangle$

label 2 means no red bond

Kitaev's toric code model



- the simplest and most well studied spin model with topological order, introduced by Alexei Kitaev

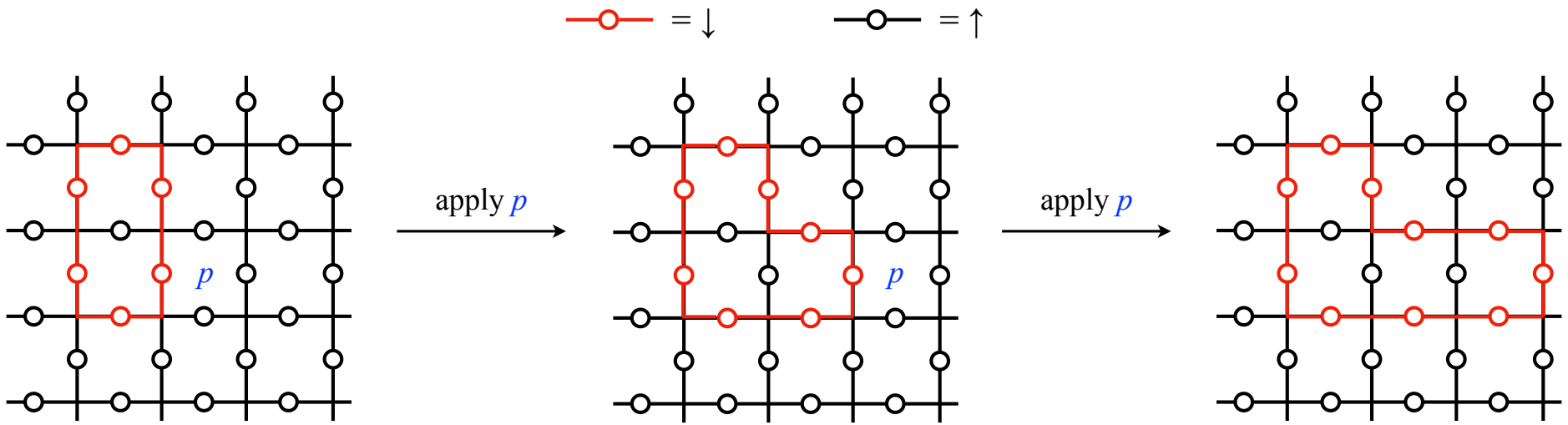


$$H = -\sum_{v} \prod_{j \in \text{vertex}} \sigma_j^z - \sum_{p} \prod_{j \in \text{plaquette}} \sigma_j^x$$

- vertex and plaquette terms commute with each other because they have either 0 or 2 common edges

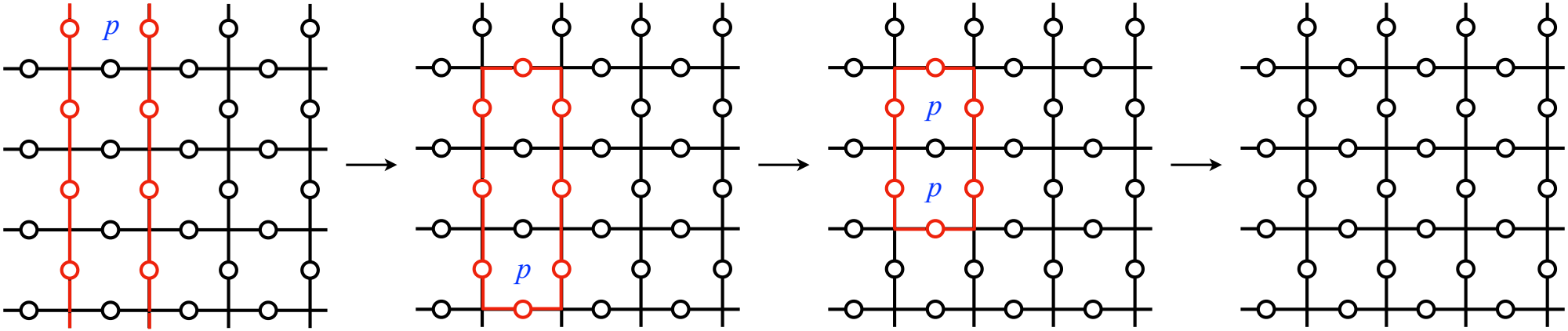
- vertex and plaquette terms have eigenvalues +1 or -1

- ground states are equal weight superposition of all closed loops

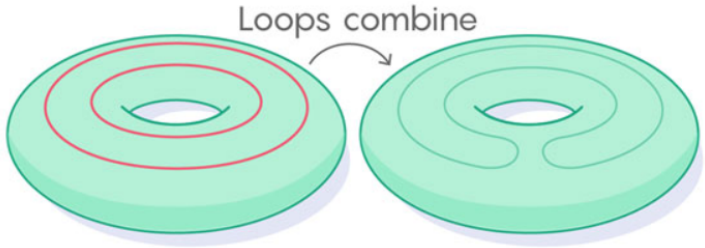


Ground state degeneracy

- topological invariance

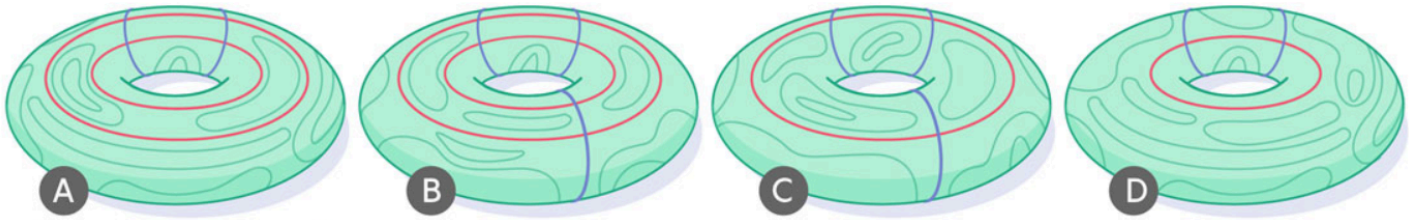


2 loops (even)



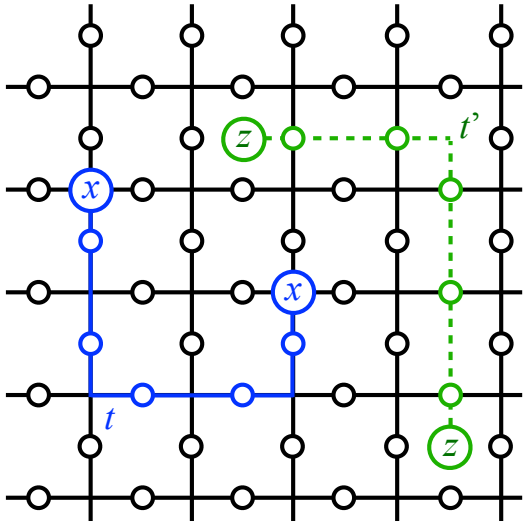
0 loops (also even)

- ground states are 4-fold degenerate on a torus



an even number of loops run around hole	even	odd	odd
an even number of loops run through hole	odd	odd	even

Abelian anyons



$$S^x(t) = \prod_{j \in t} \sigma_j^x, S^z(t') = \prod_{j \in t'} \sigma_j^z$$

- two particles are created at the end points of the string t

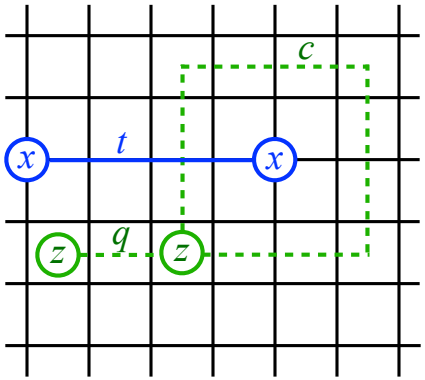
  such particles live on the vertices of the lattice

  we call them x-type particles or electric charges

- correspondingly, z-type particles or magnetic vortices are created at the end points of the string t'

  such particles live on the plaquettes

- what happens if we move particles around each other?



a z-type particle moving around a x-type particle

$$|\psi_{\text{initial}}\rangle = S^x(t)\, S^z(q)\, |G\rangle$$

$$|\psi_{\text{final}}\rangle = S^z(c)\, S^x(t)\, S^z(q)\, |G\rangle = -S^x(t)\, S^z(c)\, S^z(q)\, |G\rangle = -S^x(t)\, S^z(q)\, |G\rangle = -|\psi_{\text{initial}}\rangle$$

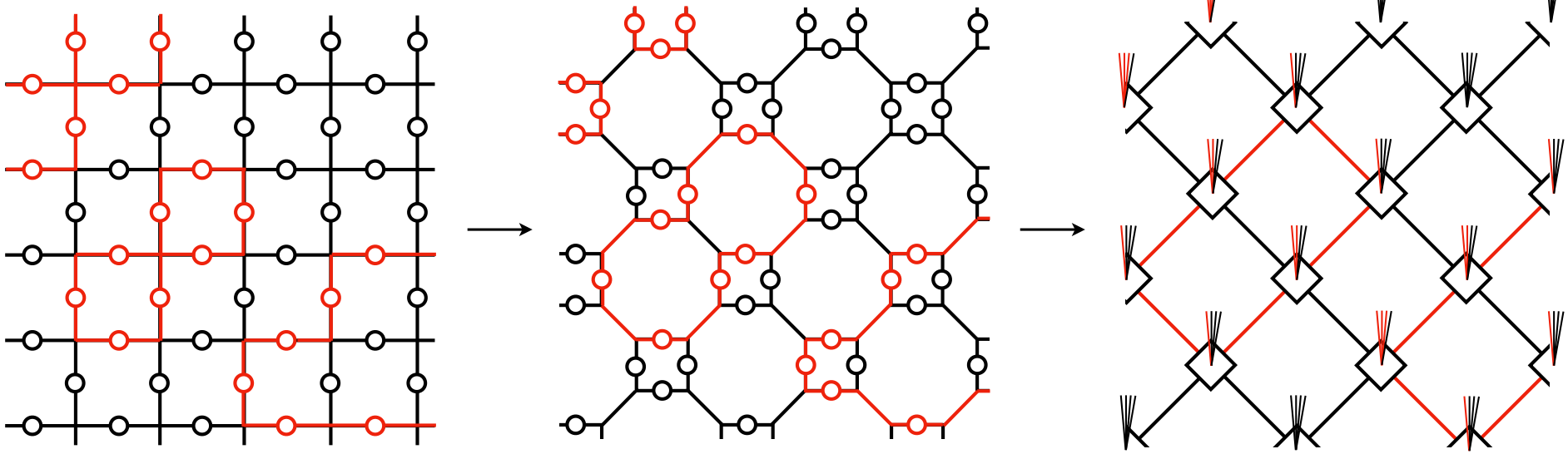
- unlike usual particles, bosons and fermions, which do not change their phase in such a process

- the global wave function acquires the phase factor -1

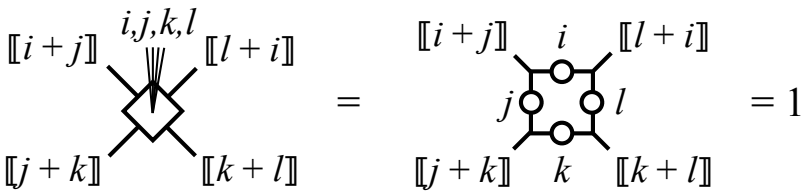
  particles with this unusual property are called abelian anyons

PEPS for toric code ground states

- the total number of  at every vertex must be even

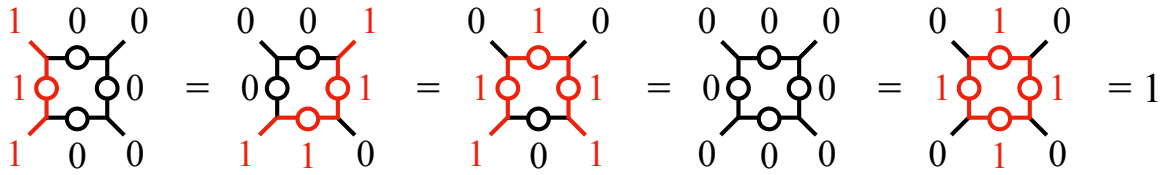


- non-trivial elements of PEPS



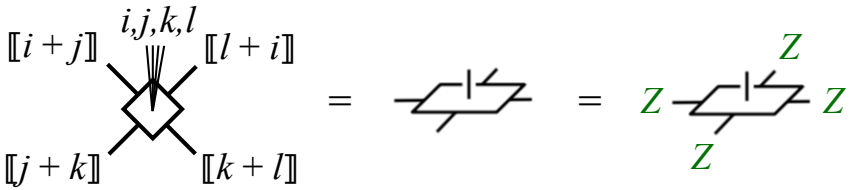
$$i, j, k, l \in \{0, 1\}$$
$$[\![i + j]\!] = \mathrm{mod}\,(i + j, 2)$$

- example



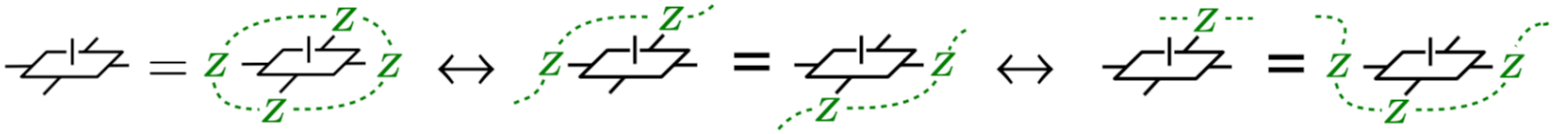
Symmetry of toric code PEPS

- toric code tensor has $\mathbb{Z}_2$ symmetry (even parity)

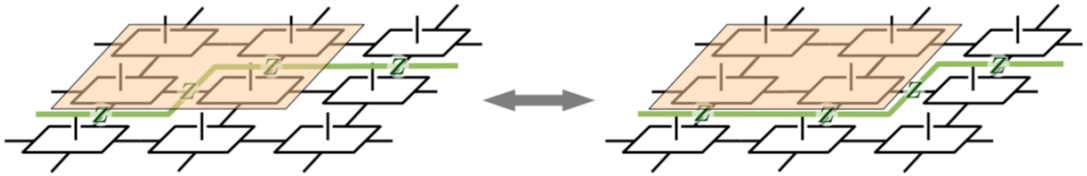


$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

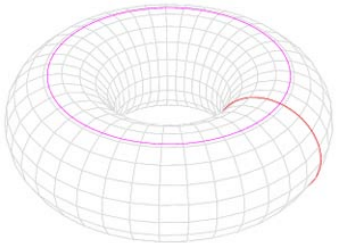
- symmetry can be rephrased as pulling through condition

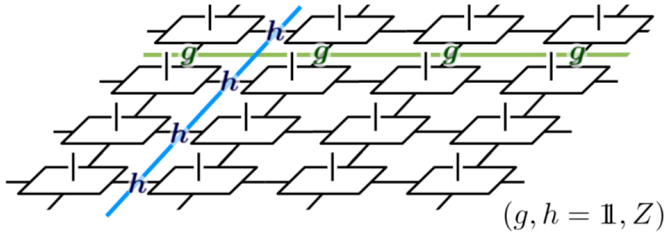


- strings can be freely moved, are invisible locally to Hamiltonian



- closed strings yield different ground states

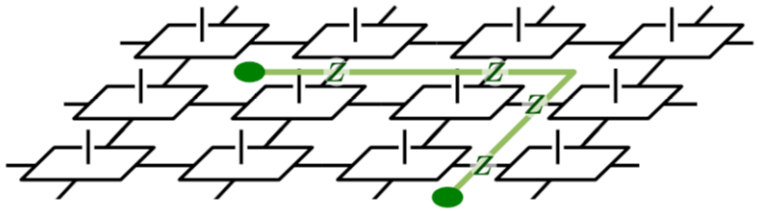




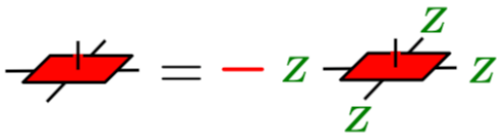
- local characterization of global topological order

Symmetries and excitations

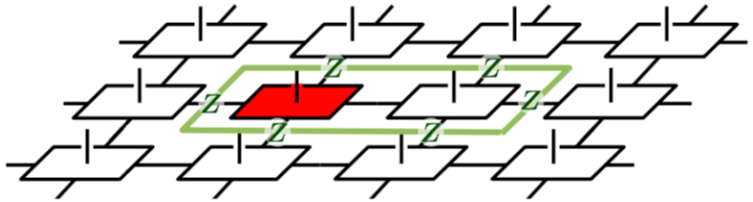
- strings with open ends correspond to excitations, must come in pairs



- tensors with odd parity correspond to excitations, must come in pairs



- these two types of excitations have non-trivial mutual statistics



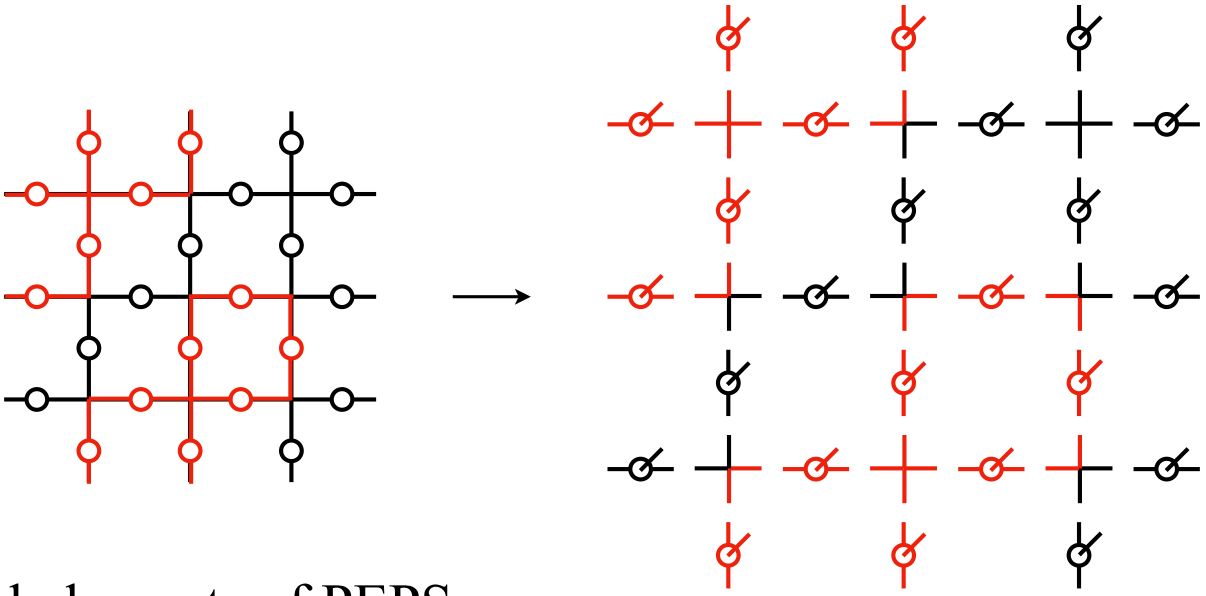
- modeling of anyonic excitations from local symmetries of tensor

  provide tools to explicitly construct and study ground states and low-energy excitations

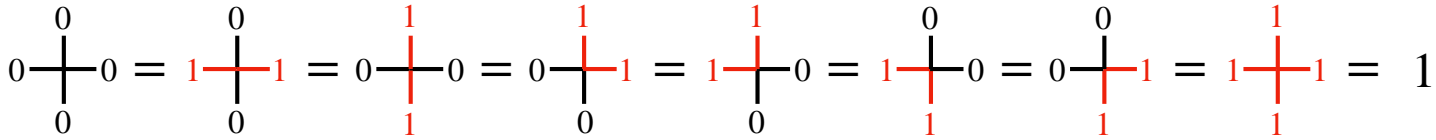
  works for systems with finite correlation length

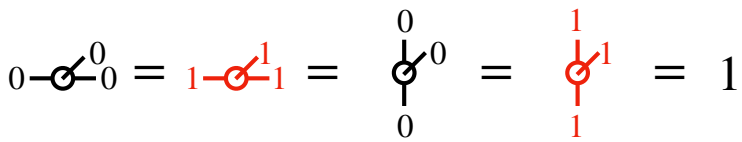
Another PEPS representation

- the total number of  at every vertex must be even

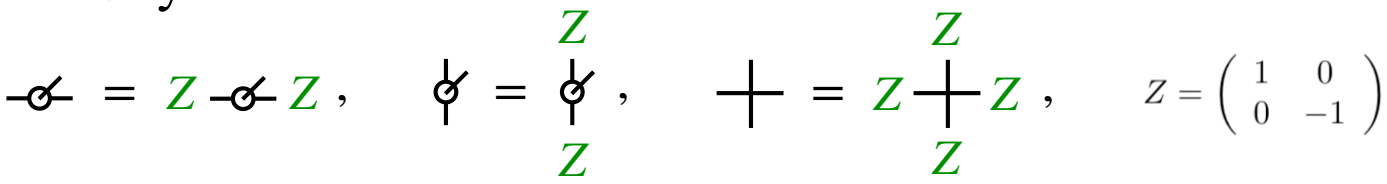


- non-trivial elements of PEPS

vertex tensor 

bond tensor 

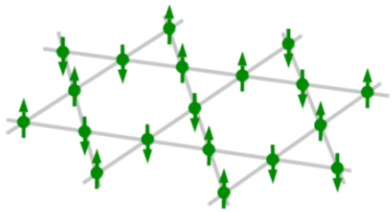
- $\mathbb{Z}_2$ symmetry

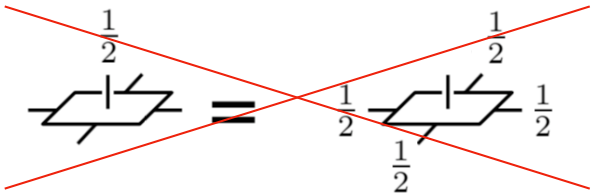


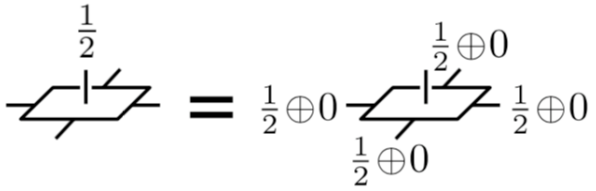
Resonating Valence Bond (RVB) states

- spin-1/2 model on the Kagome lattice

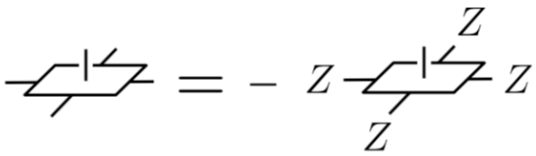
  how to encode SU(2) symmetry?

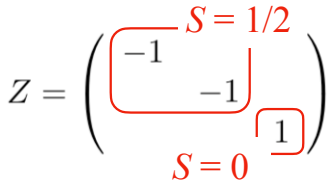






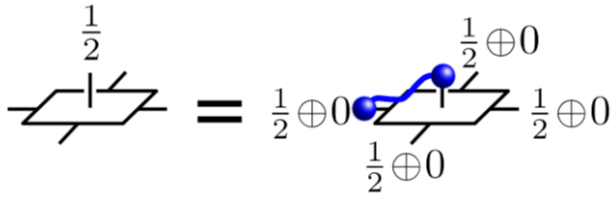
- constraint: number of half-integer representations must be odd

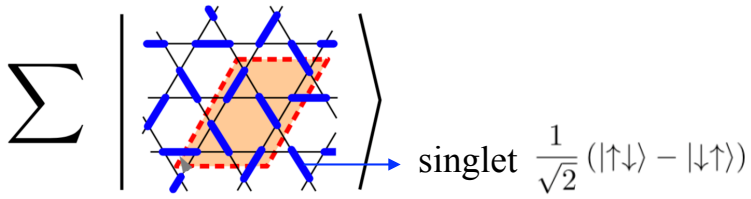




  entanglement symmetries can emerge from physical symmetries

- SU(2) invariant PEPS on the Kagome lattice





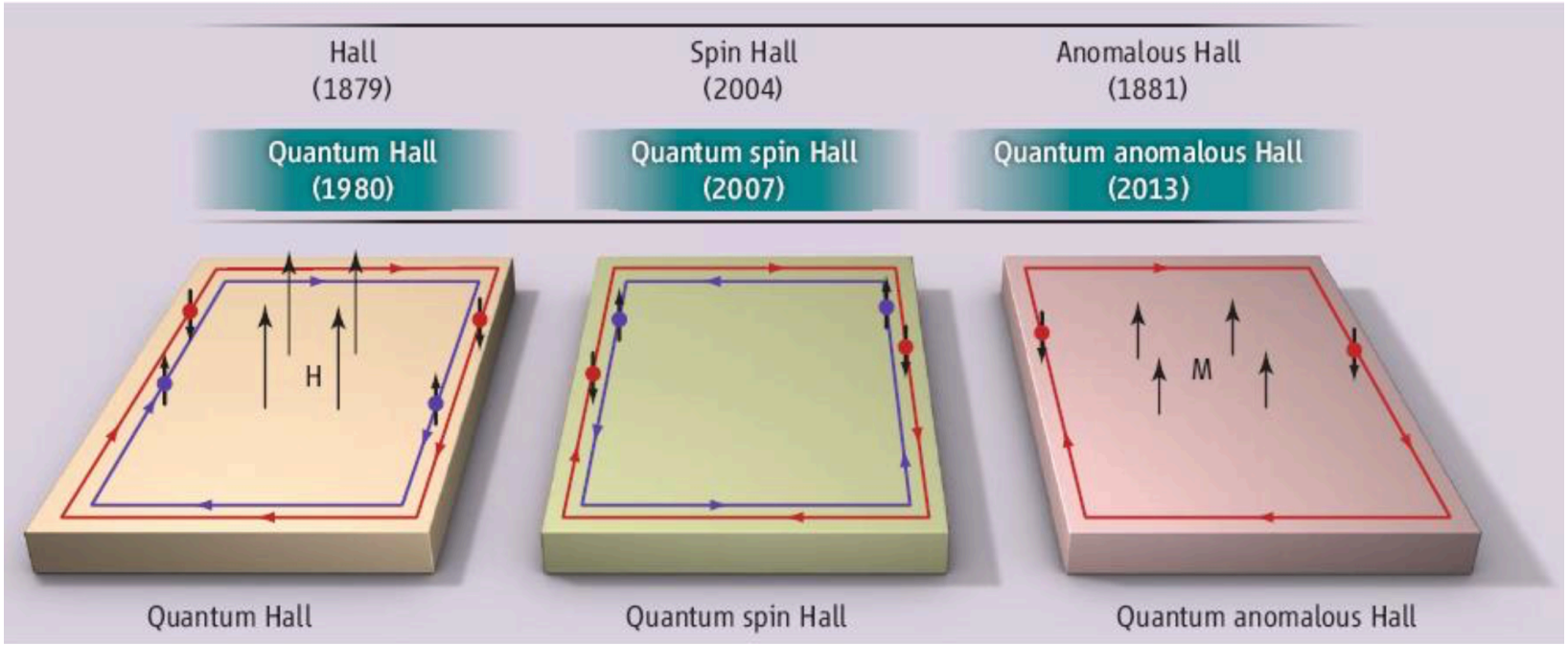
  natural interpretation of $\mathbb{Z}_2$ constraint: fixed parity of singlets along cut

Edge physics of topological models

- Fractional Quantum Hall Effect (FQHE)
  edge exhibits precisely quantized currents, robust to any perturbation

- such a behavior cannot occur in a truly one-dimensional system
  physics at the edge has an anomaly

- origin of anomalous edge physics: presence of topologically entangled bulk
  nature of anomaly characterizes topological order in the bulk

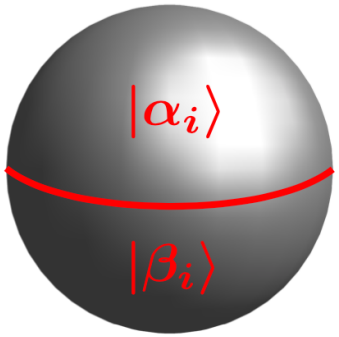


Entanglement spectrum

- entanglement spectrum

$$|\psi\rangle = \sum_i e^{-E_i} |\alpha_i\rangle \otimes |\beta_i\rangle$$

FQHE state $|\psi\rangle$

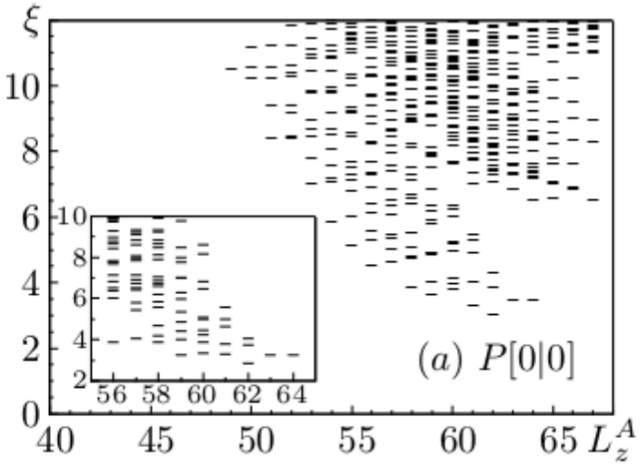


entanglement spectrum (ES)

momentum k associated to 1D boundary

→ spectrum of a 1D "entanglement Hamiltonian"

- FQHE: entanglement spectrum resembles spectrum of anomalous edge theory (a conformal field theory)



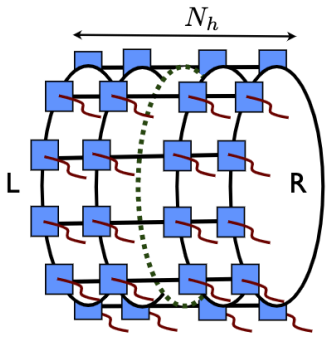
- ES can help to characterize topological phases

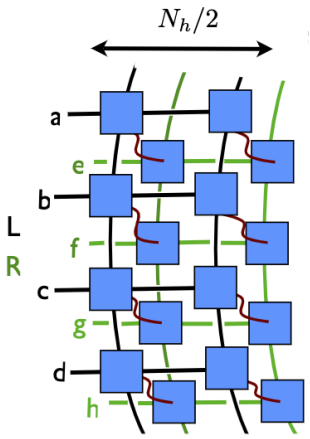
- can we understand the relation between entanglement spectrum, edge physics, and topological order in the bulk?

- can we understand why the entanglement spectrum relates to a 1D system?

Bulk-edge correspondence in PEPS

- after bipartition, entanglement carried by degrees of freedom at boundary





$$|\Psi_L\rangle = \sum_{I_a} |\hat{L}^{I_a})|I_a\rangle, \quad |\Psi_R\rangle = \sum_{I_a} |\hat{R}^{I_b})|I_b\rangle$$

$$\sigma_L = \sum_{I_a} |\hat{L}^{I_a})(\hat{L}^{I_a}| \,, \quad \sigma_R = \sum_{I_b} |\hat{R}^{I_b})(\hat{R}^{I_b}|$$

$$\rho_\ell = U\sqrt{\sigma_L^T}\sigma_R\sqrt{\sigma_L^T}U^\dagger$$

$$\sigma_L^T = \sigma_R =: \sigma_b \qquad \rho_\ell = U\sigma_b^2 U^\dagger$$

- this allows for direct derivation of entanglement Hamiltonian lives on entanglement degrees of freedom

$$\sigma_b^2 = \exp(-H_{\text{ent}})$$

- H_{ent} has natural 1D structure

  H_{ent} inherits all symmetries from tensor

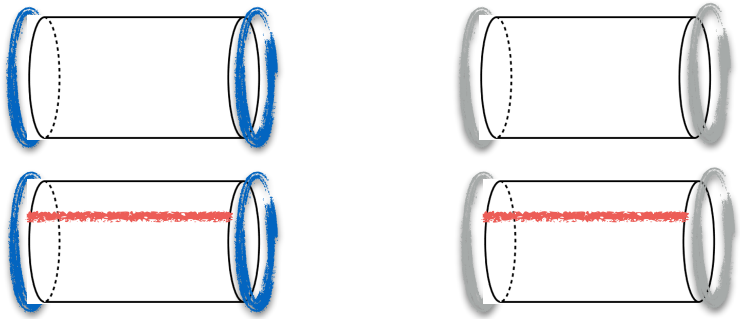
J. I. Cirac, D. Poilblanc, N. Schuch, F. Verstraete, Phys. Rev. B 83, 245134 (2011)

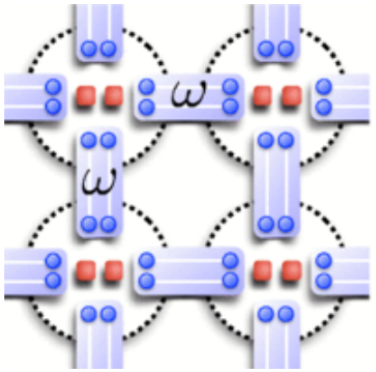
Entanglement spectrum

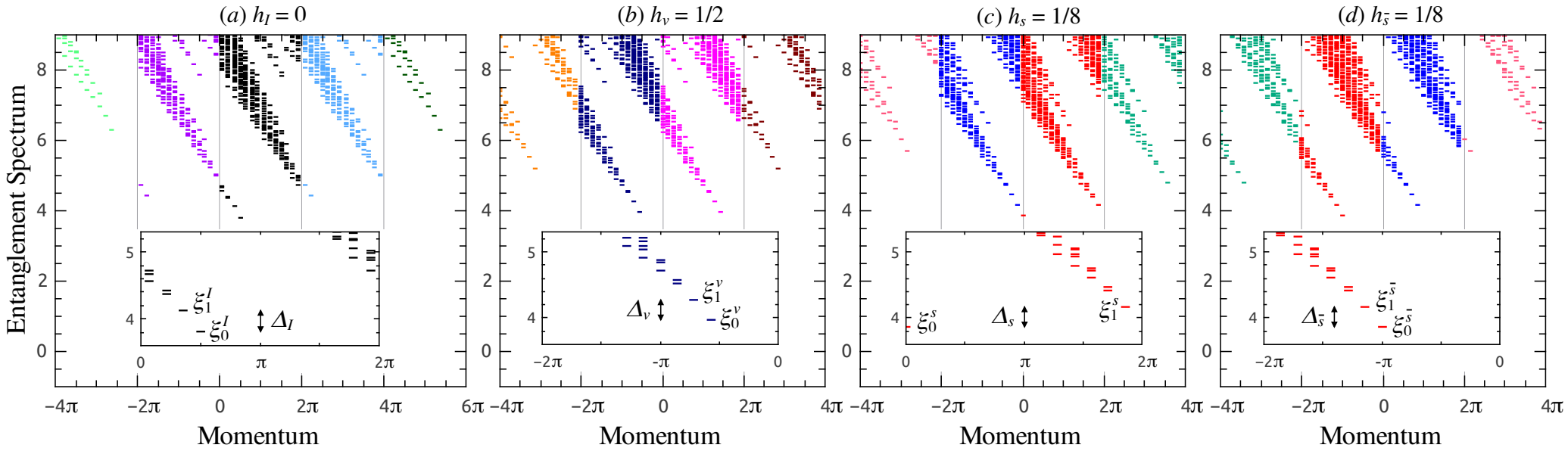
- minimally entangled states

no flux

with flux







- degeneracy pattern is {1,1,2,3,5,7, …}, $c = 1$ chiral Luttinger liquid theory

S. Yang, T. Wahl, H.-H. Tu, N. Schuch, and J.I. Cirac, Phys. Rev. Lett. 114, 106803 (2015)

Tensor networks

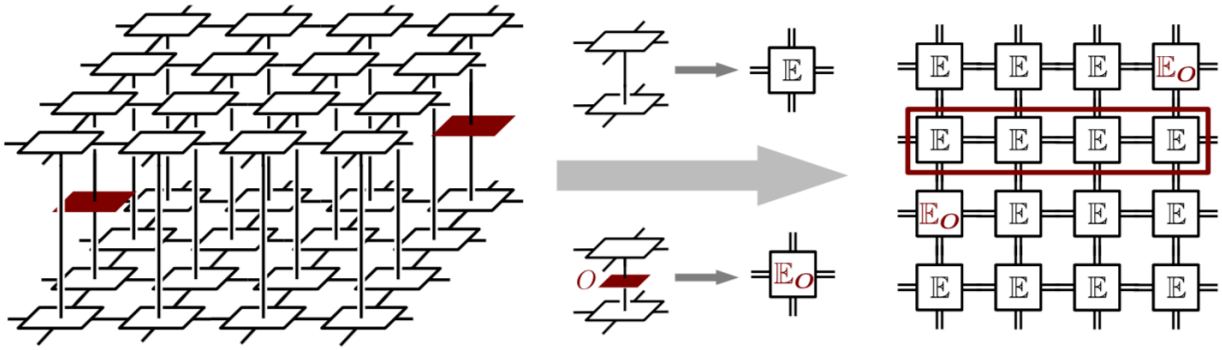
- Projected Entangled-Pair States (PEPS) and topological order
- 2D tensor network algorithms
- Multi-scale Entanglement Renormalization Ansatz (MERA)

Selected review articles:
F. Verstraete, J.I. Cirac, and V. Murg, arXiv: 0907.2796.
R. Orus, arXiv:1306.2164.
N. Schuch, arXiv:1306.5551.
J. C. Bridgeman and C. T. Chubb, arXiv:1603.03039.
R. Orus, arXiv: 1812.04011.
J.I. Cirac, D. Perez-Garcia, N. Schuch, F. Verstraete, arXiv:2011.12127.

New progress (update daily)
http://quattro.phys.sci.kobe-u.ac.jp/dmrg/condmat.html

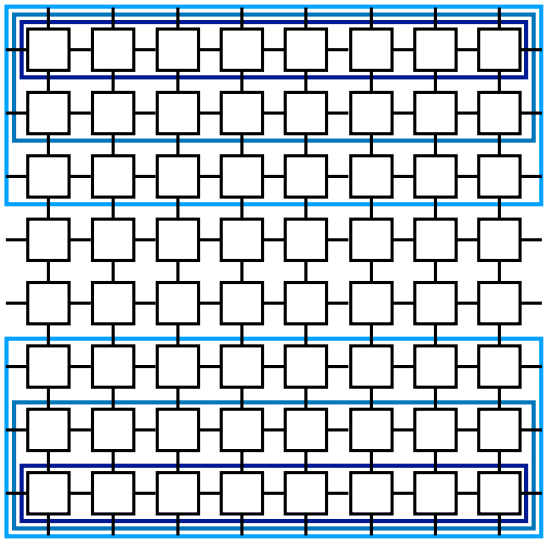
Computational complexity of PEPS

- expectation values in PEPS (e.g., correlation function)

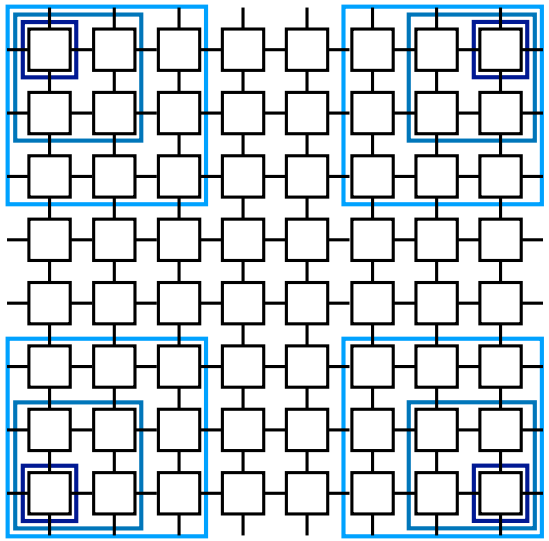


- exact contraction is a #P hard problem

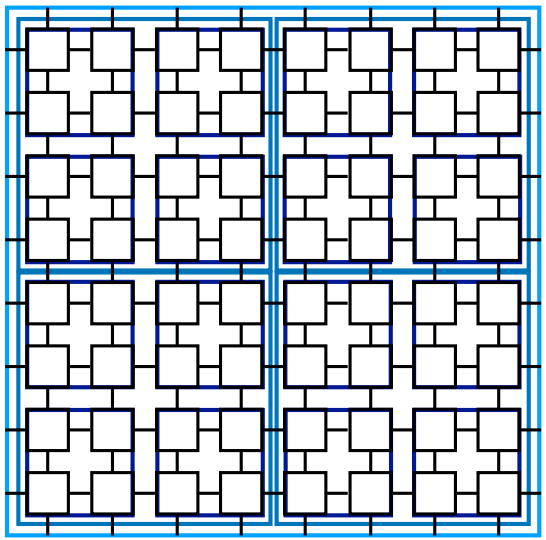
  approximation methods to contract 2D tensor networks:



boundary MPS



corner transfer matrices

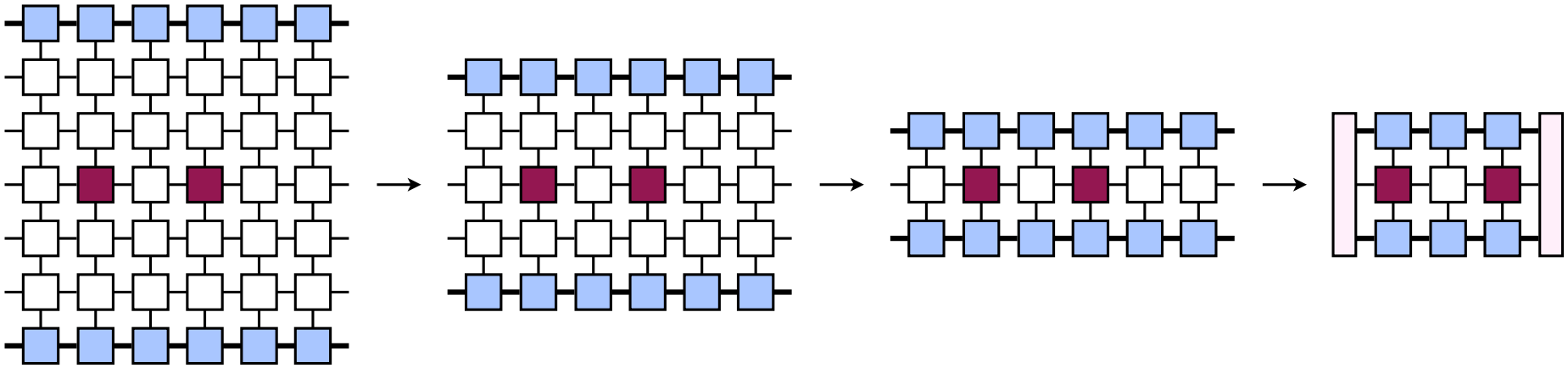


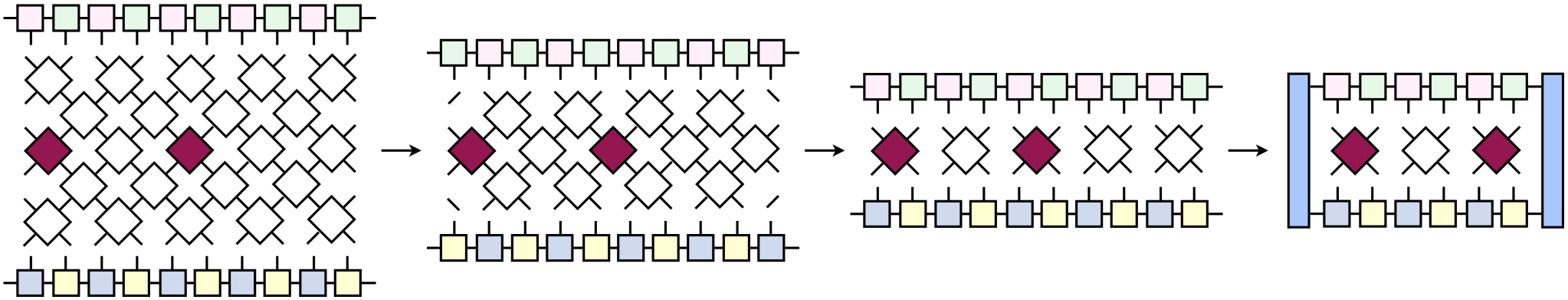
tensor coarse-graining

Boundary MPS

- contract the full 2D lattice start from a boundary

  boundary tensors forming MPS, bulk tensors forming MPO





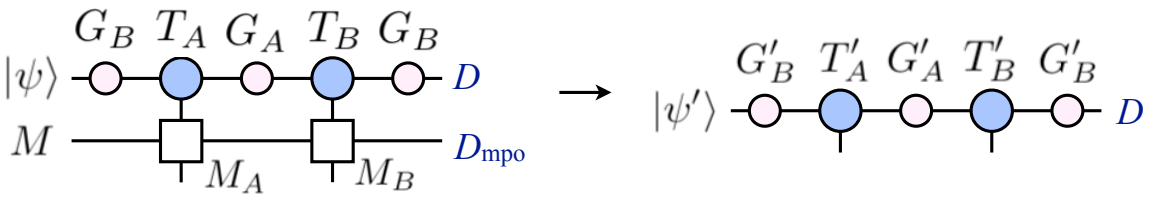
- first contract vertically using iTEBD or vMPS method (need truncation)

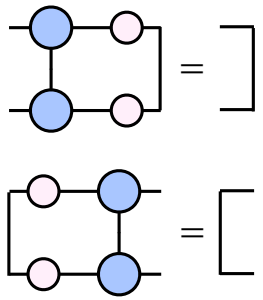
  then contract horizontally (exact contraction)

Non-unitary time evolution

- we aim to build an iMPS for the state $|\psi'\rangle = M|\psi\rangle$

  $|\psi\rangle$ and $|\psi'\rangle$ are in canonical forms

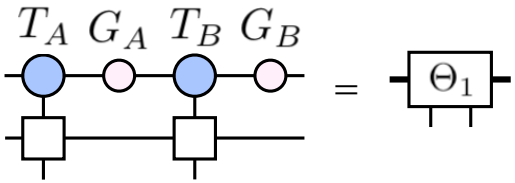




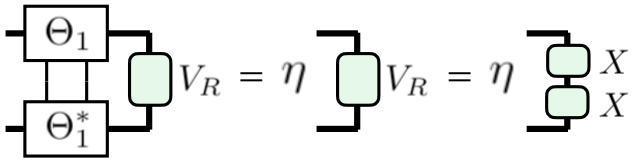
- steps to find G'_A, G'_B, T'_A, T'_B

R. Orus and G. Vidal, Phys. Rev. B 78, 155117 (2008)

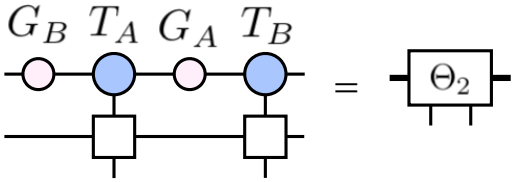
(1)



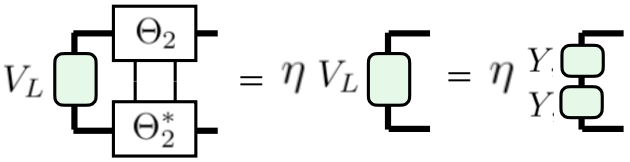
(2)



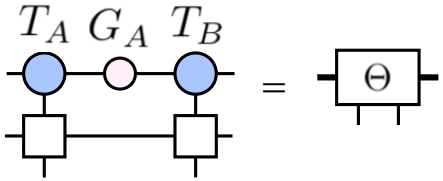
(3)



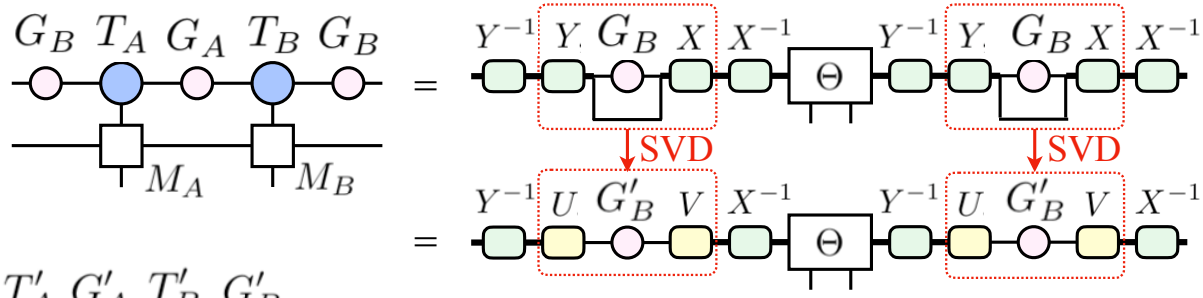
(4)



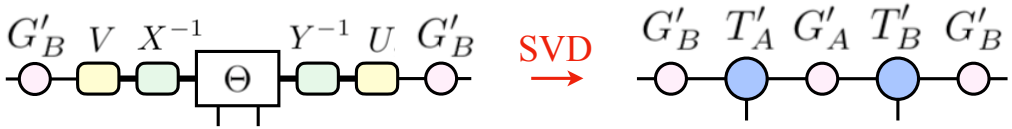
(5)



(6)

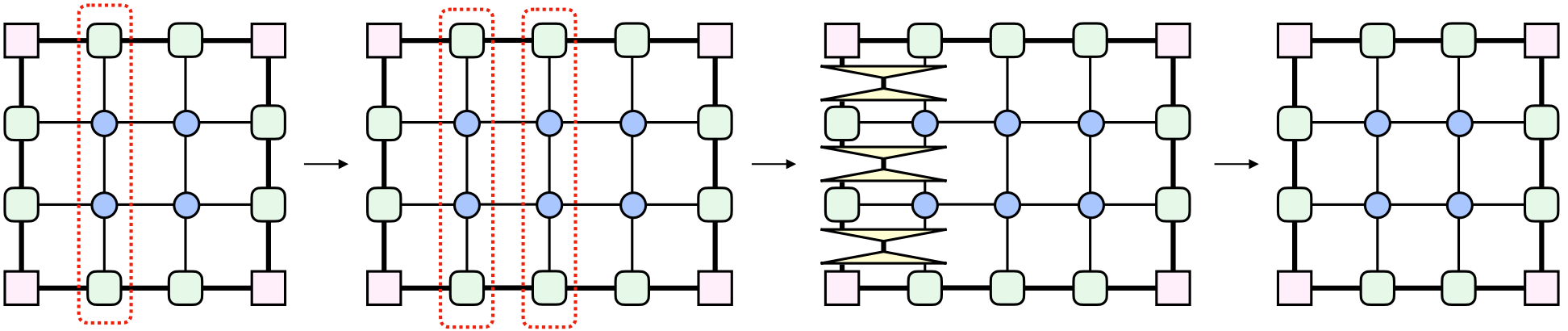


(7)



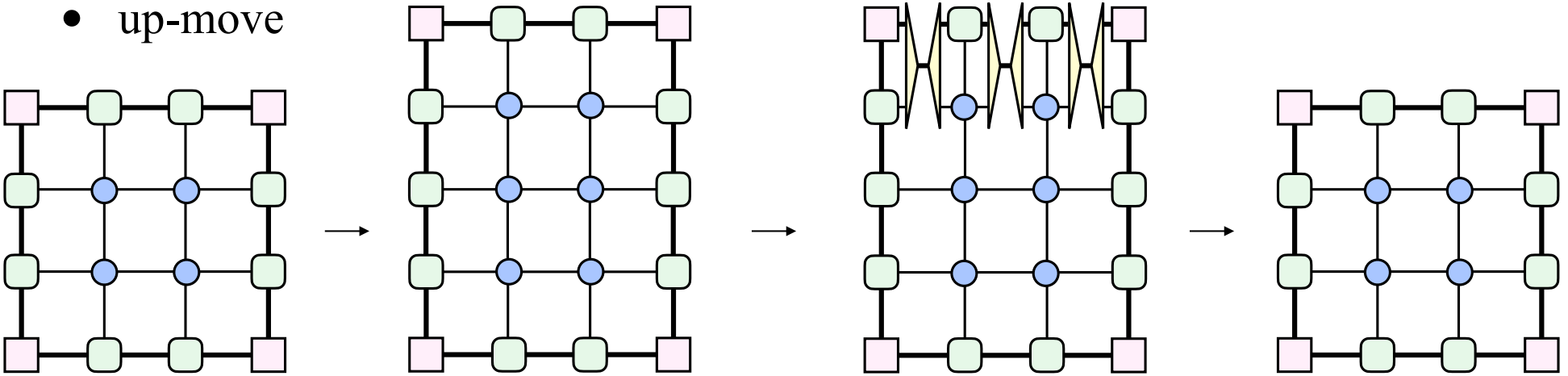
Corner transfer matrices (CTM)

- enlarge the corner by absorbing system gradually to get the fixed point
- left-move



insertion absorption renormalization

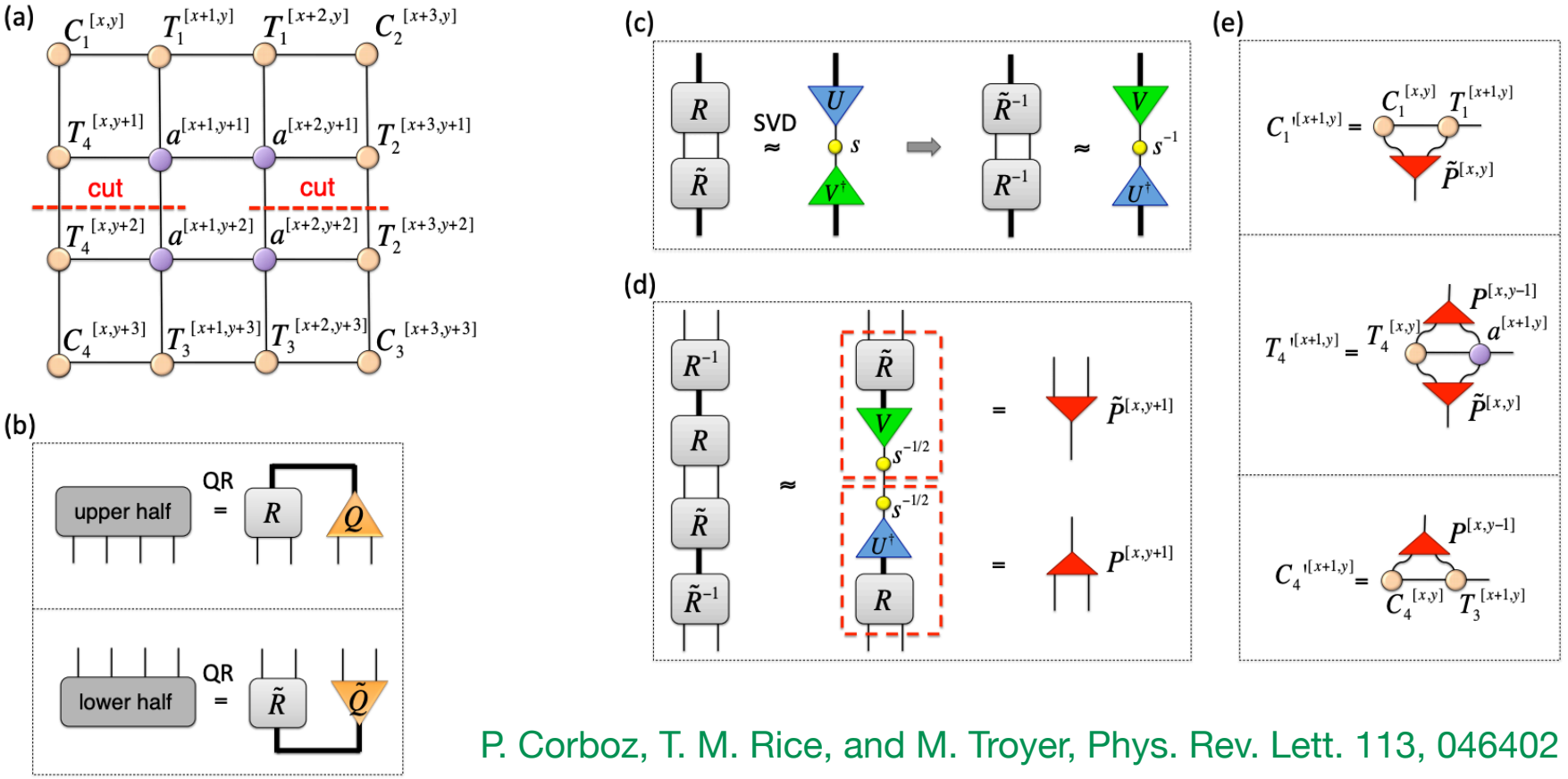
- up-move



- iterate in all directions (left, right, up, down) until convergence

R. Orus and G. Vidal, PRB 80, 094403 (2009); R. Orus, PRB 85, 205117 (2012)

Details to find the projectors



P. Corboz, T. M. Rice, and M. Troyer, Phys. Rev. Lett. 113, 046402 (2014).

(a) 4 × 4 tensors is split into an upper half and a lower half

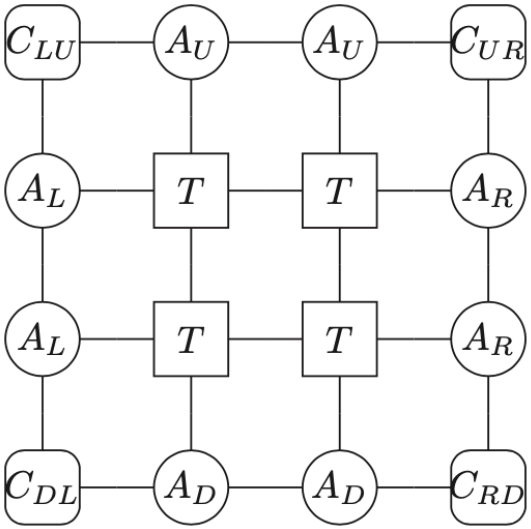
(b) QR decomposition (or SVD) is performed on the upper and lower half

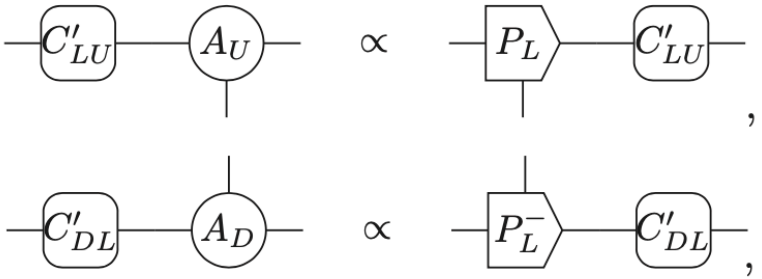
(c) SVD is performed and only partial singular values are kept

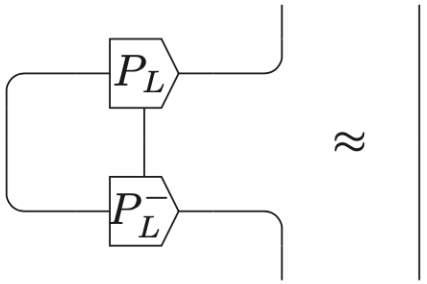
(d) identity is approximated from (c), which yields the projectors

(e) use projectors to absorb a column of tensors into the left environment tensors

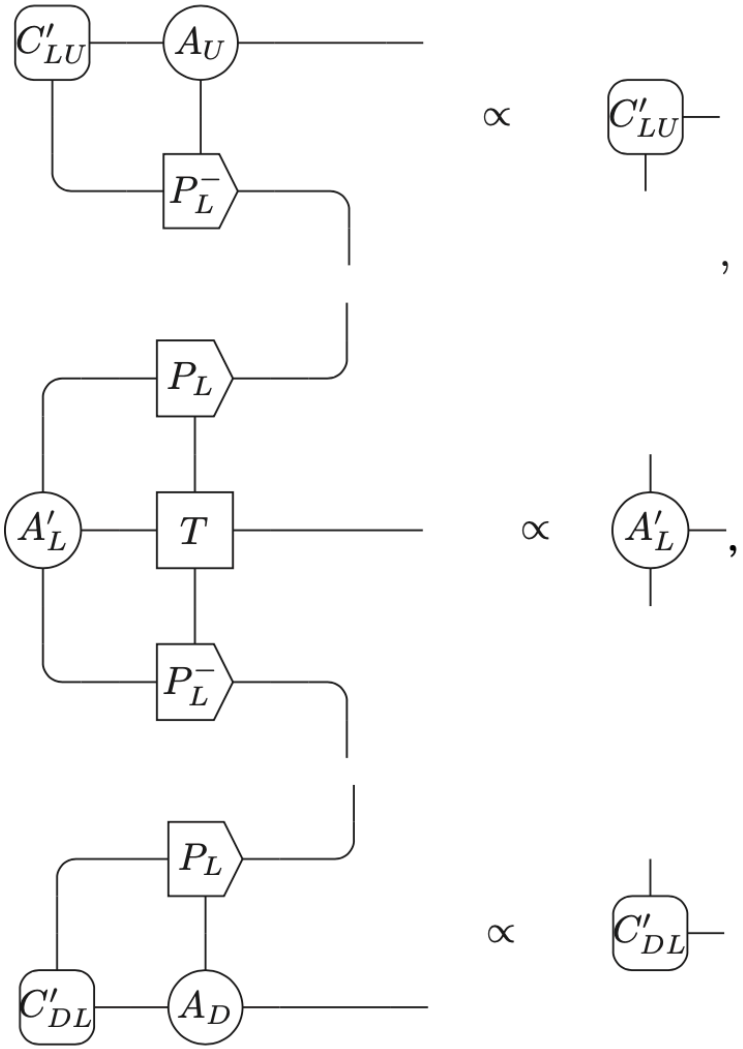
Fixed point corner method







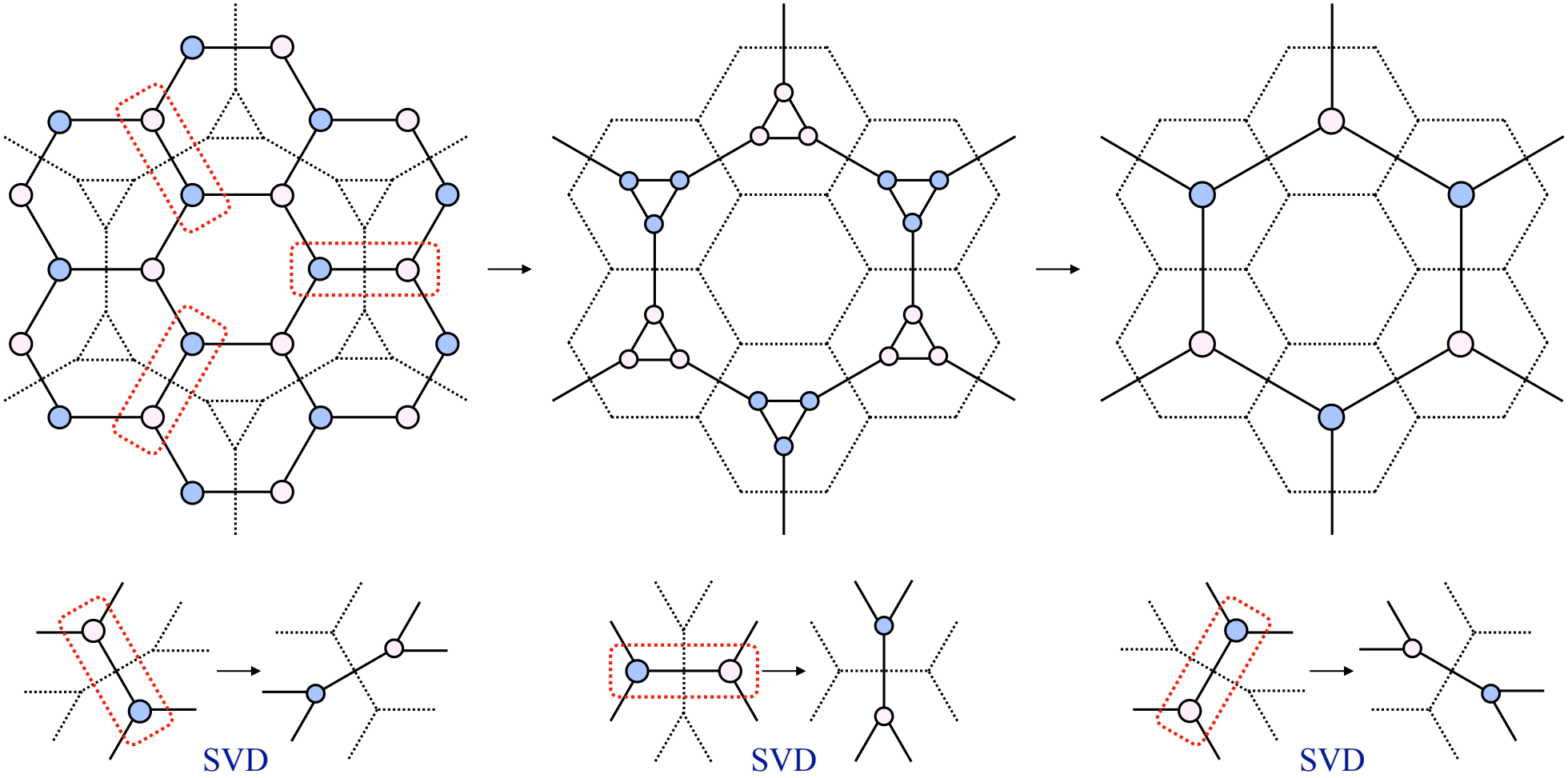
- fixed point equations



M. T. Fishman, *et. al*, Phys. Rev. B 98, 235148 (2018).

Tensor coarse-graining (TRG)

- honeycomb lattice

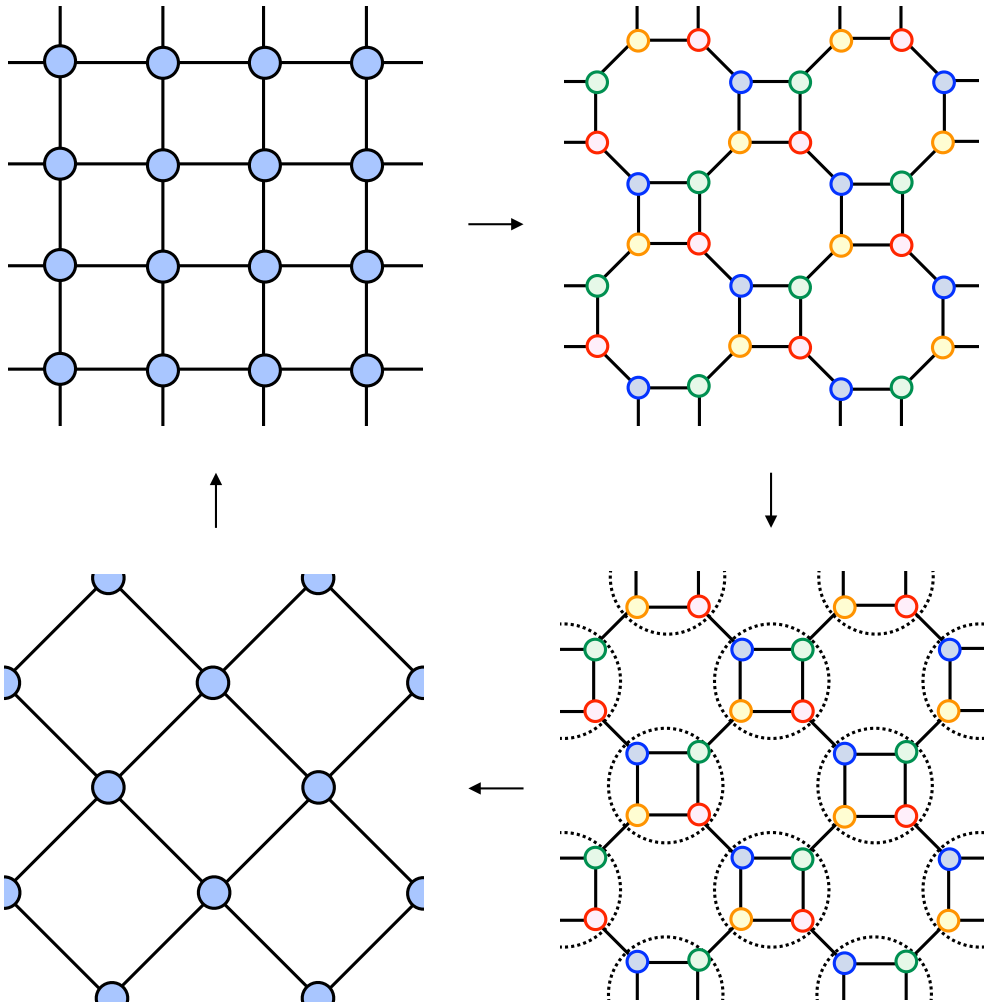


- the local degree of freedom is renormalized and decimated in each scale transformation

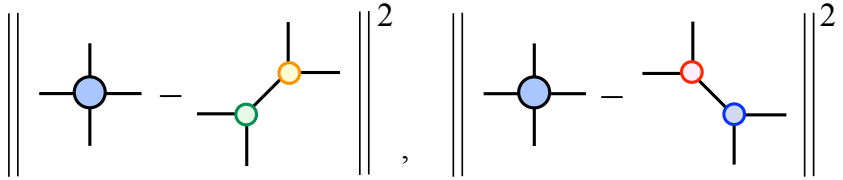
M. Levin and C. P. Nave, PRL 99, 120601 (2007)

Tensor coarse-graining (TRG)

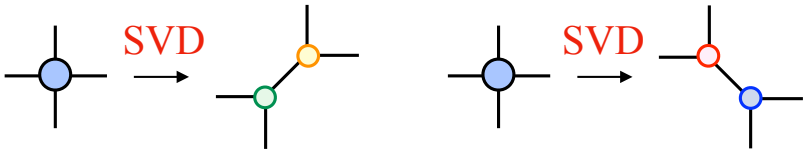
- square lattice



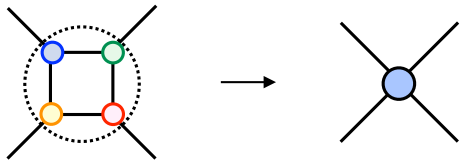
1. deform tensors, make a truncation minimizing local cost functions



which can be done by local SVD



2. coarse graining

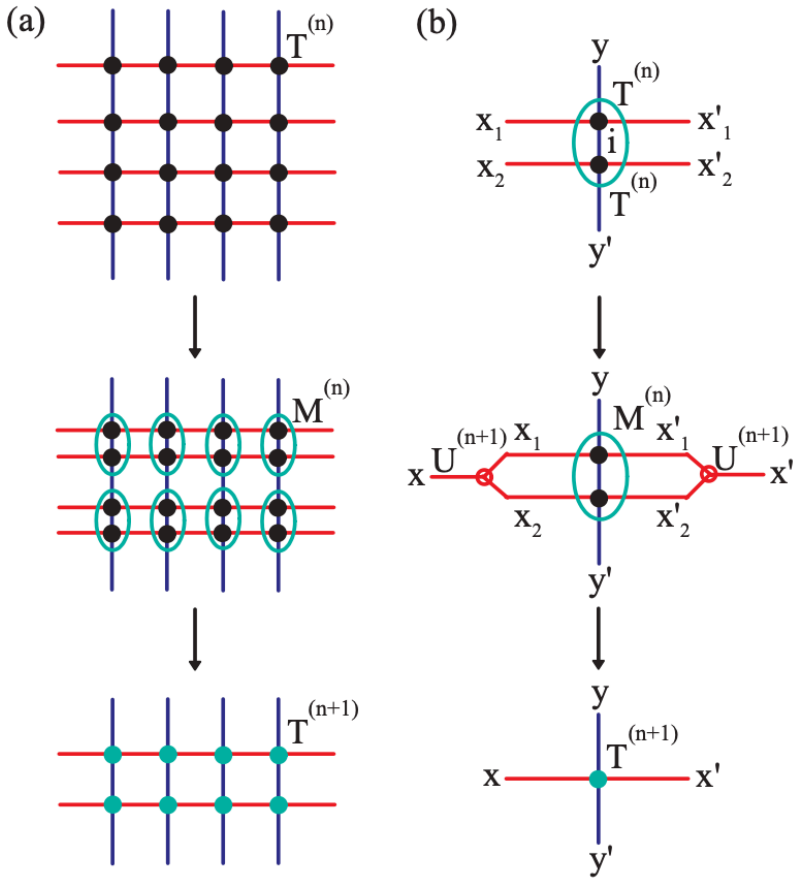


3. renormalize tensors
   multiply by a constant factor

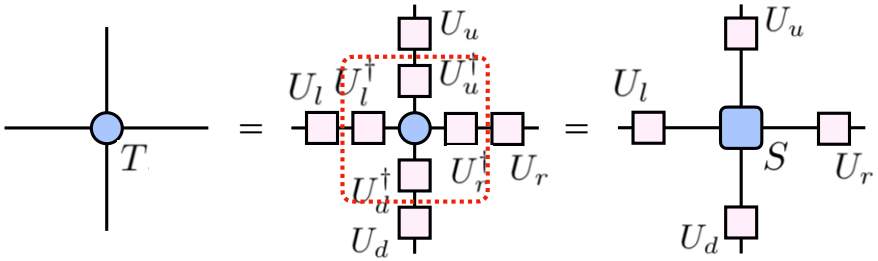
M. Levin and C. P. Nave, PRL 99, 120601 (2007)

Higher-order TRG (HOTRG)

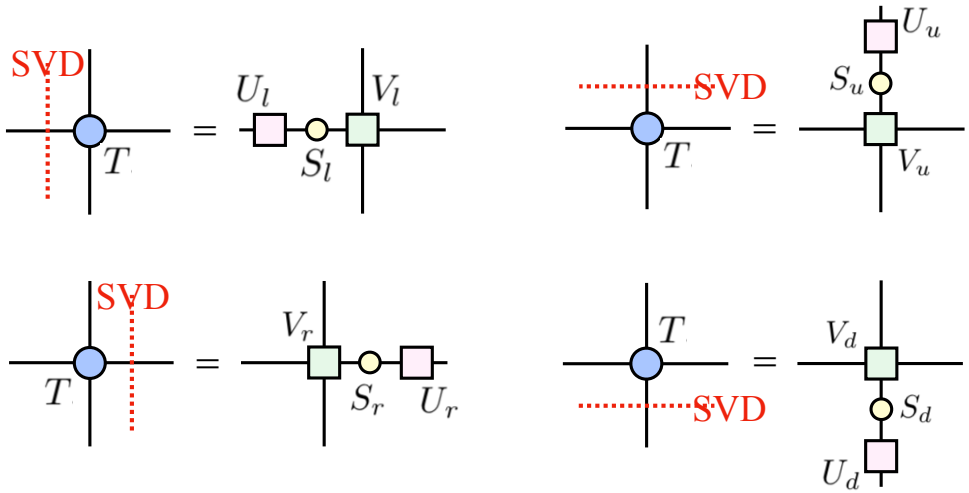
- coarse grain along the lattice vectors alternatively



- higher-order singular value decomposition (HOSVD)



- U are obtained by directional SVD independently



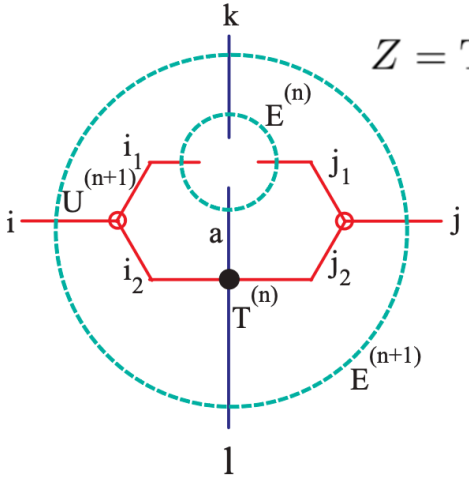
- two directions are coarse-grained alternatively to cover more entanglement and keep symmetry

Z. Y. Xie, J. Chen, M. P. Qin, J. W. Zhu, L. P. Yang, and T. Xiang, PRB 86, 045139 (2012)

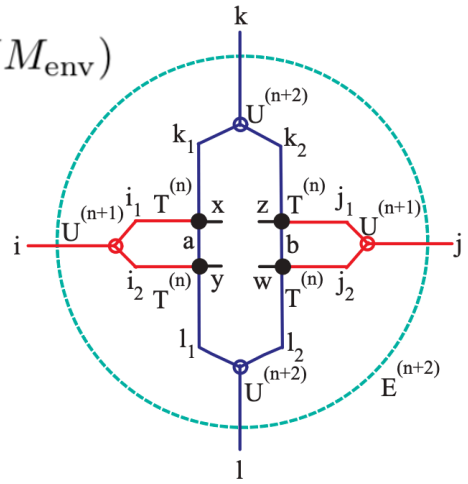
Extension of HOTRG

- HOTRG is a local update method without environment

  HOSRG: consider the environment in the frame work of HOTRG



$$Z = \mathrm{Tr}\,(MM_{\mathrm{env}})$$



- forward iteration:

  HOTRG to obtain U at all scales

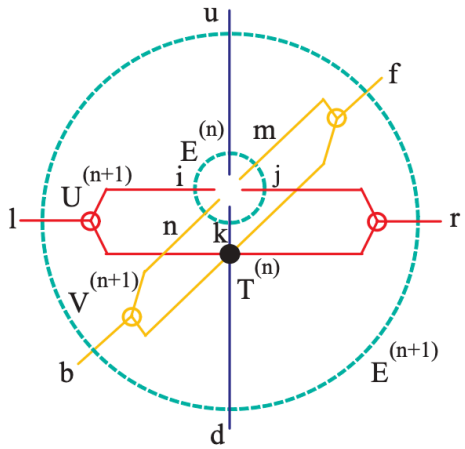
- backward iteration:

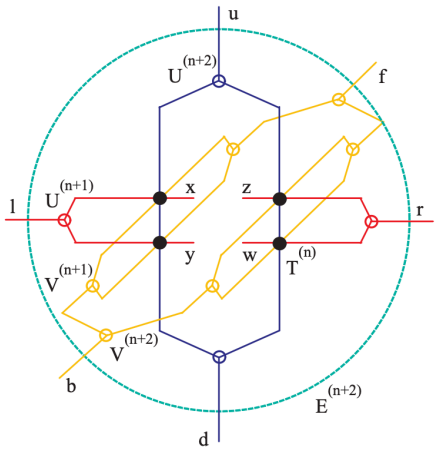
  get E from the recursive relation

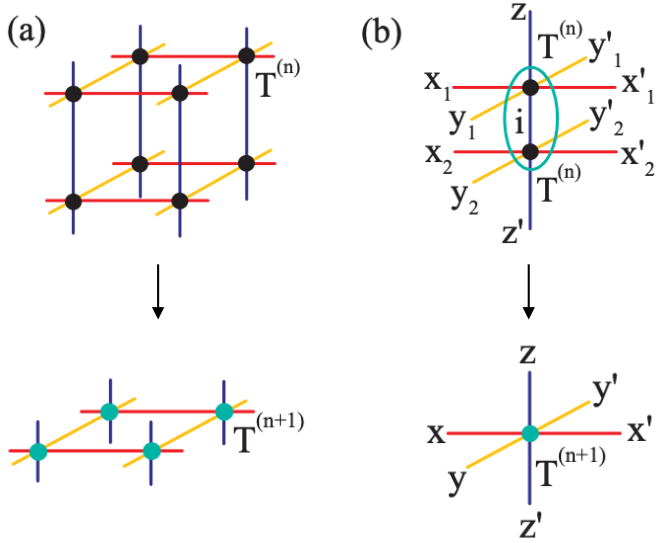
- sweep:

  repeat to gain more accuracy

- extension to 3D



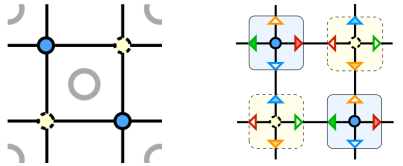
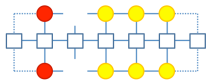
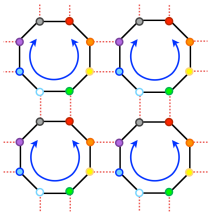
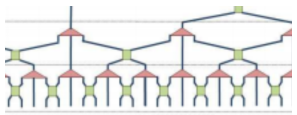
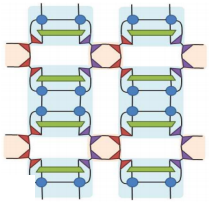




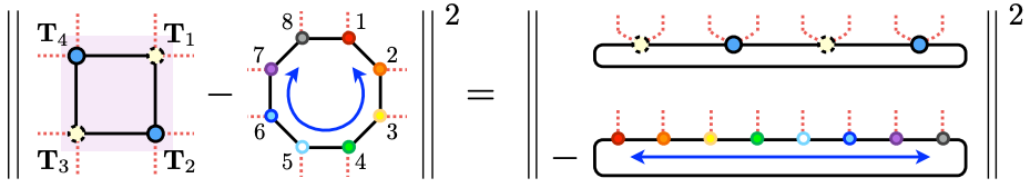
Z. Y. Xie, J. Chen, M. P. Qin, J. W. Zhu, L. P. Yang, and T. Xiang, PRB 86, 045139 (2012)

Improvement of TRG

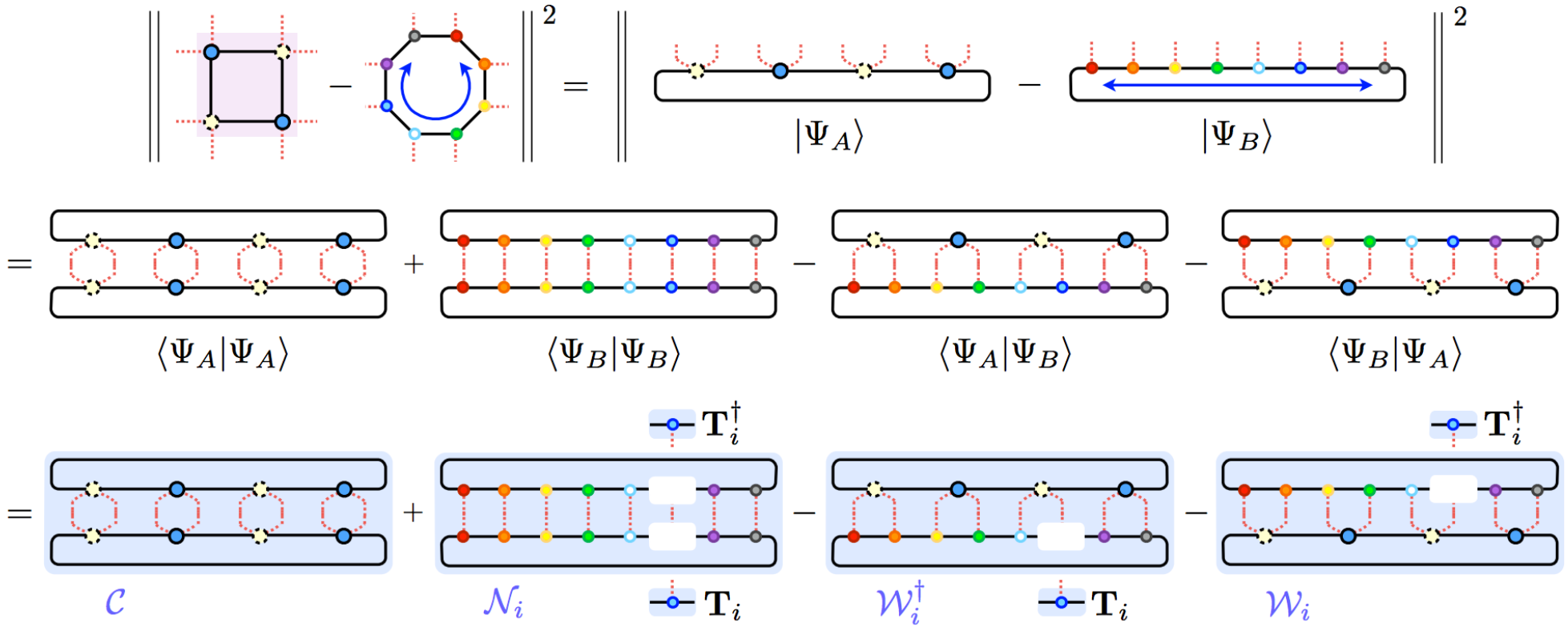
- integrate 1D algorithms with TRG to enhance its performance

1D algorithm	TRG + 1D algorithm
 iTEBD	GW-TNR Z.-C. Gu and X.-G. Wen, PRB 80, 155131 (2009) 
 DMRG Varational MPS	Loop-TNR S. Yang, Z.-C. Gu, X.-G. Wen, PRL 118, 110504 (2017) 
 MERA	EV-TNR G. Evenbly and G. Vidal, PRL 115, 180405 (2015) 

- loop-TNR



Key step in Loop TNR

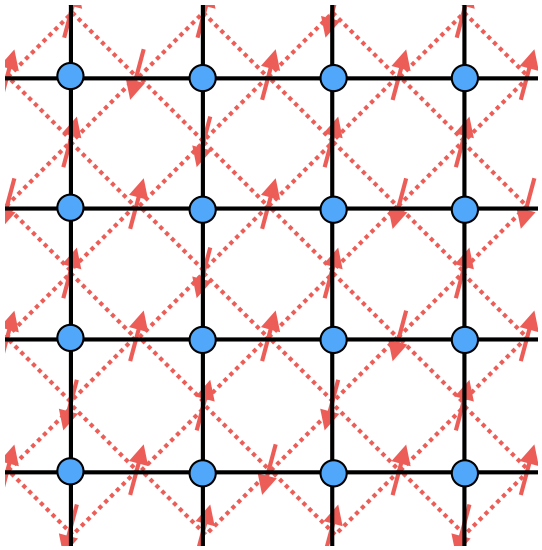


$$f(\mathbf{T}_i) = \||\Psi_A\rangle - |\Psi_B\rangle\| = \langle\Psi_A|\Psi_A\rangle + \langle\Psi_B|\Psi_B\rangle - \langle\Psi_A|\Psi_B\rangle - \langle\Psi_B|\Psi_A\rangle$$
$$= \mathcal{C} + \mathbf{T}_i^\dagger \mathcal{N}_i \mathbf{T}_i - \mathcal{W}_i^\dagger \mathbf{T}_i - \mathbf{T}_i^\dagger \mathcal{W}_i,$$

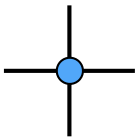
- solve the linear equation $\mathcal{N}_i \mathbf{T}_i = \mathcal{W}_i.$

S. Yang, Z.-C. Gu, X.-G. Wen, PRL 118, 110504 (2017)

Example: classical Ising model



- partition function $Z = \exp(-\beta H) = \sum_{\{\sigma\}} \exp(\beta \sum_{\langle ij \rangle} \sigma_i \sigma_j)$
- local tensor



$$\mathbf{T} = \sum_{u,l,d,r} T^{\text{Ising}}_{u,l,d,r}$$

$$T^{\text{Ising}}_{0,1,0,1} = e^{-4\beta}, \quad T^{\text{Ising}}_{1,0,1,0} = e^{-4\beta},$$

$$T^{\text{Ising}}_{1,1,1,1} = e^{4\beta}, \quad T^{\text{Ising}}_{0,0,0,0} = e^{4\beta},$$

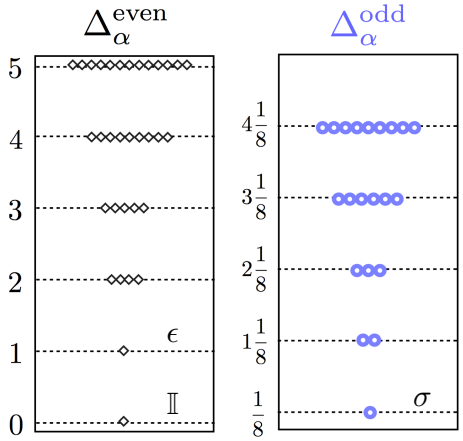
others $= 1$

- Ising CFT

central charge

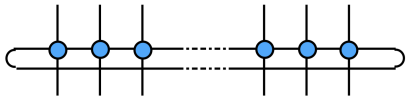
$c = 1/2$

scaling dimensions $\longrightarrow$



- calculate scaling dimensions

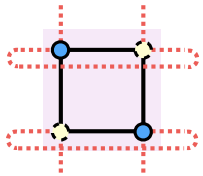
transfer matrix



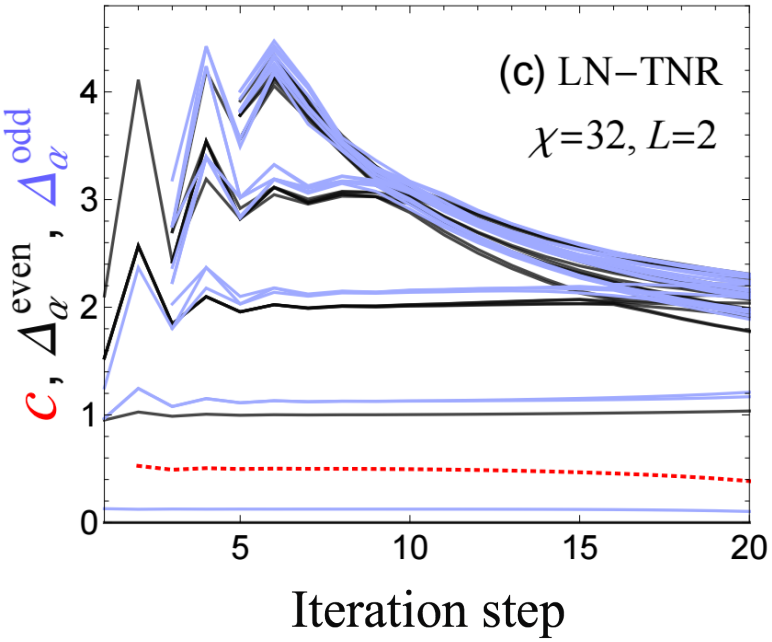
eigenvalues of transfer matrix $\longrightarrow$ c, Δ_α

Z.-G. Gu and X.-G. Wen, PRB 80, 155131 (2009).

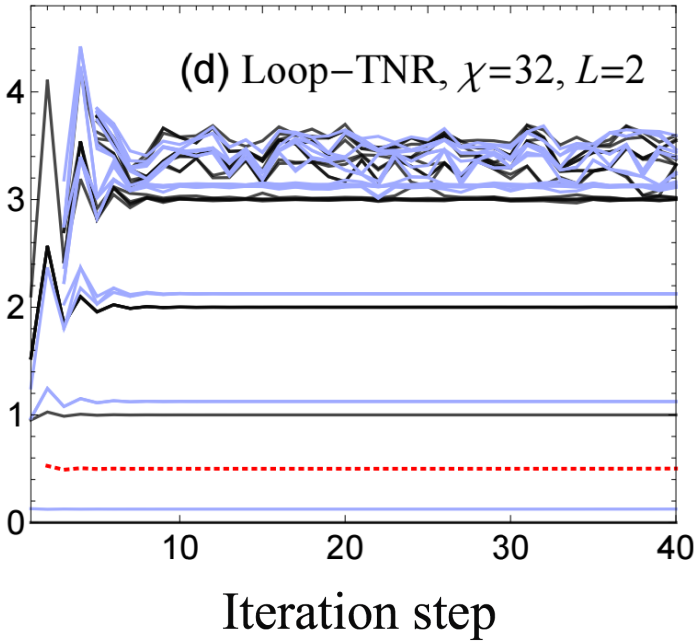
Comparison of results



- LN-TNR results



- Loop-TNR results



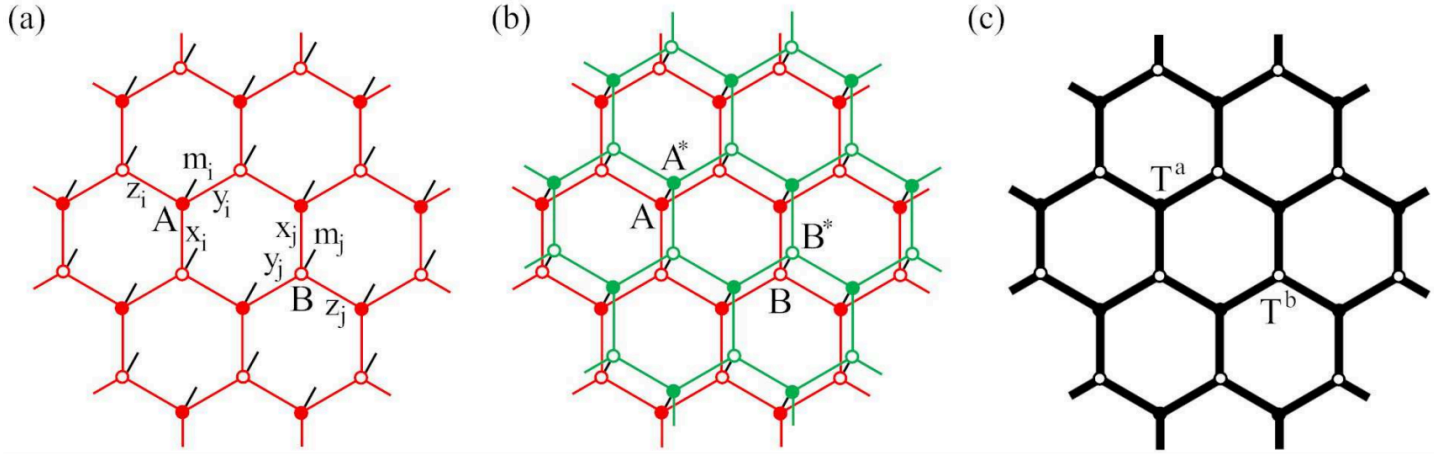
- scaling dimensions change with scale ~ cannot recover scale invariance
- high-index parts will destroy low-index parts

- scaling dimensions does not change with scale ~ scale invariance
- a clear gap between high-level parts and low-level parts

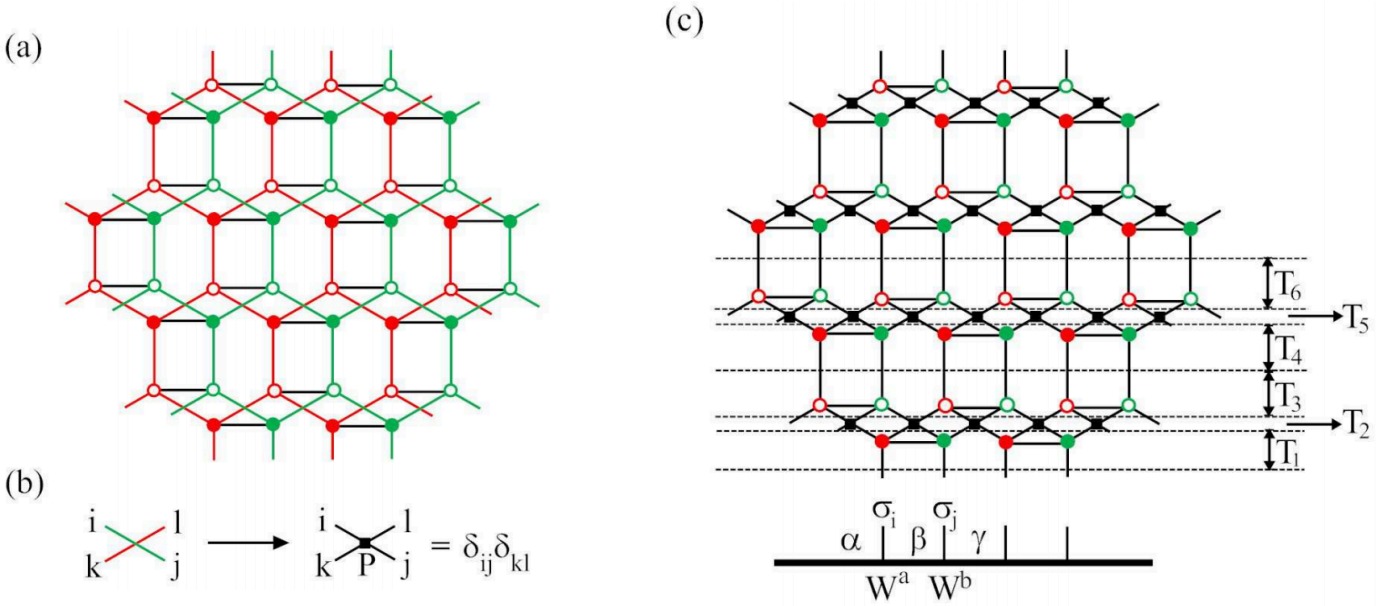
S. Yang, Z.-C. Gu, X.-G. Wen, PRL 118, 110504 (2017)

Nested tensor network

- double-layer tensor network representation of $\langle\psi\,|\psi\rangle$



- nested single-layer mapping of the double-layer tensor-network state



Z. Y. Xie, H. J. Liao, R. Z. Huang, H. D. Xie, J. Chen, Z. Y. Liu, and T. Xiang, PRB 96, 045128 (2017)

New progress: TRG in 3D

PHYSICAL REVIEW E **111**, 054140 (2025)

Three-dimensional real-space renormalization group with well-controlled approximations

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We make Kadanoff's block idea into a reliable three-dimensional (3D) real-space renormalization group (RG) method. Kadanoff's idea, expressed in spin representation, offers a qualitative intuition for clarifying scaling behavior in criticality, but has difficulty as a quantitative tool due to uncontrolled approximations. A tensor-network reformulation equips the block idea with a measure of RG errors. In 3D, we propose an entanglement filtering scheme to enhance such a block-tensor map, with the lattice reflection symmetry exploited. When the proposed RG is applied to the cubic-lattice Ising model, the RG errors are reduced to about 2% by retaining more couplings. The estimated scaling dimensions of the two relevant fields have errors 0.4% and 0.1% in the best case, compared with the accepted values. The proposed RG is promising as a systematically improvable real-space RG method in 3D. The unique feature of our method is its ability to numerically obtain a 3D critical fixed point in a high-dimensional tensor space. A fixed-point tensor contains much more information than a handful of observables estimated in conventional techniques for analyzing critical systems.

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